

Program 3 – Doubly Linked List

Assignment Objective: Build skills on C++ class creation while implementing a **doubly linked list of key-value pairs**.

Requirements:

- As with P1 List, a dllTemplate.cpp file is provided; it must be copied to dll.cpp. The formatting and ordering therein shall be maintained.
- In files called dll.h and dll.cpp, do the following.
- In dll.h, include common.h as it had been in programs 1 and 2.
- Create a class called **node** with the following members:
 - Private members:
 - KEY_VALUE kv;
 - node *next, *prev;
 - Public:
 - constructor ... node(KEY_VALUE kv, node *next = NULL, node *prev = NULL)
 - Friend dll
- Create a class called **dll** with the following members
 - Private members:
 - node *first, *last;
 - unsigned int listCount; // to record the number of entries in the list
 - unsigned int getIndex(int key, node *pn, int index) const; // recursion helper
 - void printIt(node *pn, unsigned int index, int limit) const; // recursion helper
 - void printItBackwards(node *pn, unsigned int index, int limit) const; // recursion helper
 - void clear(node *pn); // recursion helper
 - Public members:
 - dll() // causes the object to be initialized (first=last= NULL and nCount = 0).
 - dll(dll* srcList); // causes a new list to be returned that is a complete copy of the srcList.
 - ~dll() // deletes the dynamically allocated nodes in the list.
 - Create the same member functions that were in list and cList, with the following modifications:
 - The functionality of these functions shall be via recursion: clear(), getIndex(), printIt(), printItBackwards().
 - getCapacity() always returns the same as getCount().
- Create a Makefile per classroom instructions that makes each .o file and the final program, named dll. This shall use the provided listMain.cpp file. Use the -D DO_PARTB option when compiling listMain.cpp to be listMain.o.
- Demonstrate the correctness of the implementation by doing the following, using the listMain.cpp that is given to you on D2L:
 - Just do: make
 - Execute the code and inspect the results; if they are what you expect from your understanding of listMain.cpp, proceed to the submission; otherwise, inspect and fix your code as necessary and repeat these steps.
 - The correct output is shown in the D2L file dllCorrectOutput.txt.
- **Submission process:**
 - Create the file dllOutput.txt using the following command in your environment:
 - ./dll > dllOutput.txt
 - **Printing:**

- Your dll.h and dll.cpp shall be printed with line numbers, and with a constant pitch font; no variable pitch fonts will be accepted. Ensure that none of your code and/or comment lines wrap to a second line.
 - Submit your printouts in the following order: dll.h, dll.cpp, and dllOutput.txt.
 - NO screenshots will be accepted.
 - Bring the printouts to class when due.
- o **D2L submission:**
- Submit a single zip file to D2L with the Makefile, dll.h, dll.cpp, and dllOutput.txt files.